Supplement A - Technical Details for The Link Function Problem in Psychological Interaction Testing

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S1. Scope of this supplement

This supplement contains extended review methods and reliability, the chance-corrected-link algebra, the generative parameters behind the illustrative figures of the main paper, full simulation specifications and estimation details for the three worked simulations and for the diagnostic simulation, and fine-grained supplementary results.

S2. Preregistered review: extended methods and reliability

Protocol and sampling frame

The review protocol was preregistered before any record was screened and is available at https://osf.io/bdu73/, together with the coding data dictionary. The protocol specified a descriptive review of recent psychological research articles from five pre-specified journals, one per major area: *Psychological Science* (general), *Journal of Experimental Psychology: General* (experimental), *Journal of Personality and Social Psychology* (social), *Developmental Science* (developmental), and *Psychological Medicine* (clinical). Journals were selected on a discretionary combination of criteria stated in the protocol: first-quartile Impact Factor in the 2023 Journal Citation Reports, not primarily publishing reviews, not narrowly specialized, and strong general reputation in the respective field, as jointly evaluated by the coauthors.

Article records were retrieved from Elsevier's Scopus database for all articles published in these journals in 2024. For each journal, the full record list was randomly ordered and then screened in that order until 40 eligible interaction-testing articles were included, for a planned total of 200 interaction-testing articles across the five journals. Screening a larger set also allowed estimating the proportion of eligible empirical articles that test interactions. The design is targeted and descriptive: it characterizes reporting practice in influential outlets during the sampled year. It is not a random or stratified sample of all psychology journals, and the estimates are not prevalence estimates for the discipline.

Eligibility and coding workflow

Eligible articles were empirical articles presenting original psychological research. The following were excluded, per protocol: reviews and meta-analyses; editorials, commentaries, opinion papers, and theoretical articles without empirical data; qualitative-only articles; replication-only studies; and psychometric validation studies. Articles were coded in detail only if they tested at least one interaction, through statistical modeling or ANOVA. The coding scheme targeted observed-outcome analyses; articles whose relevant tests were conducted only at the latent-variable level, or whose empirical analyses relied only on structural equation models, were excluded from detailed coding.

The resulting screening flow is reported in Table 1. The table is generated from the saved review-descriptives output, so that rerendering the supplement uses the current repository values.

Table 1
Screening flow for the preregistered review, generated from the saved review-descriptives output.

Step	N	basis
records retrieved across journal sheets	2175	non-empty rows across all screening workbook sheets (one per journal)
eligible empirical articles in final dataset	331	Eligible == 1 in final review dataset
eligible articles testing at least one interaction	200	Tests_interactions == 1 in eligible rows
eligible articles not testing interactions	131	eligible minus interaction-testing
note-based SEM / latent-variable-only exclusions	12	note-based keyword search

Each interaction-testing article was independently coded by two researchers. Discrepancies were resolved through discussion or, if needed, by consulting the other coauthors. Generative AI tools were used only as an optional aid to locate relevant passages within an article; all coding used in the review is human. The preregistered variable definitions are reproduced below.

Table 2
Coded variables and their preregistered definitions.

Variable	Definition (preregistered)	Type
Tests interactions	Whether the article tests at least one interaction term in a statistical model or using an ANOVA.	Boolean
Uses non identity link function	Whether at least one tested interaction is presumably analyzed with a non-identity link function (for example, because a generalized linear model was used).	Boolean
Explicit link function	Whether the link function is explicitly declared for at least one statistical model in the article. Indicating only which generalized linear model or family was used is not enough.	Boolean
Incorrect identity link function	Whether at least one tested interaction is analyzed using an identity link function when another, non-identity link function would be more appropriate.	Boolean
Finds significant interactions	Whether at least one interaction, among those where the link was coded as incorrectly identity, yields a statistically significant result, or is considered and discussed analogously in non-frequentist analyses.	Boolean
Response variable types	Non-exclusive list of response-variable types on which interactions are tested (for example, sum score, accuracy, error count, response times, ordinal/Likert).	Category

Beyond these definitions, the coders applied a small set of operational rules for the recurring borderline decisions. Table 3 records the ones that most affect the reported counts.

Table 3

Operational rules the coders applied for the decisions that most affect the reported counts, condensed from the coding sheet. The preregistered variable definitions are in Table 2.

Variable	Coded "Yes" when	Decisions that most affect the count
Non-identity link used	at least one tested interaction is fitted with a model implying a non-identity link, that is, any GLM or GLMM beyond the Gaussian-identity default (ANOVA, ordinary regression, linear mixed model).	A model named "logistic", "probit", "Poisson", "negative binomial", "ordinal"/"cumulative", or "Gamma" regression implies a non-identity link.
Explicit link function	the link is identifiable from the report, either stated in words or named through a link-specific model.	"Logistic regression" (logit), "probit regression" (probit), and "Poisson" or "log-linear" (log) count as explicit. Naming only a family, a "generalized linear model", or a "GLMM" does not.
Formally incorrect identity-link analysis	an interaction is tested with an identity-link model (ANOVA, ordinary regression, linear mixed model, or a test on cell means) applied to an outcome with formal constraints: bounded, discrete, positive-only, or with a known chance level.	Flagged only when the constraint is clear from the report. Not flagged when the analysis is explicitly justified as an observed-score estimand, or when the outcome is plausibly continuous and far from any bound.
Significant interaction	at least one interaction already flagged as incorrect identity-link is reported as statistically significant, or is interpreted analogously (for example, a credible interval excluding zero) in a Bayesian analysis.	Restricted to interactions already flagged in the row above.
Response-variable types	—	Non-exclusive: one article can contribute several types (sum score, accuracy or proportion, error count, response time, ordinal/Likert, neural or physiological, difference score, correlation).

Two coding notes bound the interpretation of these variables. First, *Incorrect identity link function* is a conservative screening flag for a formal mismatch between the observed-outcome constraints and the identity-link generative model. Cases were flagged only when the constraint was identifiable from the published report; when the report did not allow this judgment, the case was not counted. The flag is not an error rate, not a severity grade, and does not imply that the substantive conclusion is false, since an observed-score estimand can be a deliberate and defensible target. Second, *Response variable types* is non-exclusive: one article can contribute to several categories, so the category counts do not sum to the number of articles.

Coding relied only on what a reader can observe in a published report: the analysis described, the outcome described, and any explicit statement of family or link. The review therefore documents reporting practice, not the correctness of substantive conclusions. For the same reason, the review did not classify interactions as removable or nonremovable at the article level, because published reports rarely provide the target contrast, the complete coefficients, and the assumed measurement scale that such a classification requires.

Interrater agreement

Double-coded records with complete coder pairs are summarized in Table 4, with raw percentage agreement and Cohen's kappa reported together. Several coded variables had imbalanced marginal distributions, a setting in which kappa can be conservative, so the coder-specific yes rates are reported alongside to let readers evaluate the agreement pattern directly.

Table 4
Interrater agreement for double-coded records.

Item	N	Agreement	Kappa	Coder 1 yes	Coder 2 yes
Uses non-identity link	199	92.0%	0.81	28.1%	30.2%
Explicit link reported	199	88.9%	0.66	20.6%	20.6%
Identity link flagged	197	88.3%	0.71	71.6%	72.1%
Significant flagged interaction	141	91.5%	0.68	86.5%	82.3%

By-journal descriptives

Table 5 reports the review descriptives by journal. Percentages are computed within interaction-testing articles, except for the interaction-testing percentage, which is computed within eligible articles. The main paper reports only the range of these quantities across journals and the position of *Psychological Medicine*, which had the lowest screening proportion and the highest non-identity-link proportion in this sample.

Table 5
By-journal review descriptives.

Journal	Eligible	Interaction	Interaction / eligible	Non-identity link	Explicit link	Identity flagged
Developmental Science	63	40	63.5%	25.0%	17.5%	57.5%
JEP: General	47	40	85.1%	27.5%	27.5%	75.0%
JPSP	55	40	72.7%	25.0%	15.0%	92.5%
Psychological Medicine	105	40	38.1%	40.0%	35.0%	40.0%
Psychological Science	61	40	65.6%	27.5%	20.0%	95.0%

S3. The chance-corrected binomial link

For binomial accuracy with a known chance level c, the chance-corrected link maps the linear predictor onto the interval from c to 1,

$$p_i = c + (1 - c)\, F(\eta_i),$$

where F is a cumulative distribution function, taken to be logistic in the analyses reported here. The corresponding link applies

F^{-1} to the above-chance quantity,

$$g(p_i) = F^{-1}\left(\frac{p_i - c}{1 - c}\right).$$

For a two-alternative forced-choice or true/false task with $c = .50$ this becomes $p_i = .50 + .50\,F(\eta_i)$. A standard binomial logit or probit is the special case $c = 0$. Both specifications respect the binomial sampling process, but they define different no-interaction baselines: a product term under the standard link tests additivity over the full 0-to-1 range, whereas under the chance-corrected link it tests additivity on a performance-above-chance scale that already encodes the task-implied lower asymptote.

The fixed-chance form used in the paper is the simplest member of a broader psychometric-function family. A more general version adds a lapse parameter λ that lowers the upper asymptote,

$$p_i = c + (1 - c - \lambda)\,F(\eta_i),$$

so that performance saturates below 1 when occasional lapses are plausible (Knoblauch & Maloney, 2012; Wichmann & Hill, 2001). The analyses in the paper fix $\lambda = 0$ and fix c by task design, for transparency and identifiability; the lapse-rate version is the natural extension when the design and data can support estimation of an upper asymptote.

S4. Data-generating processes and estimation details

This section gives the formal generative models and the estimation internals for the simulations. The main paper describes the purpose and interpretation of each simulation; here we state the equations, the fixed parameter values, and the fitting procedures needed to reproduce the reported behaviour. Unless noted otherwise, each scenario used $B = 300$ Monte Carlo replications with a significance level of $\alpha = .05$, and each simulation set a fixed random seed.

Generative parameters of the illustrative figures

Two figures in the main paper are illustrations rather than simulation results, so their parameters are recorded here.

The introductory count example samples $N = 250$ children with age drawn uniformly between 6 and 10 and group assigned by a fair coin. Error counts are Poisson with a log link,

$$\log \mu_i = 2.60 - 0.55\,(\text{age}_i - 6) + 0.65\,\text{group}_i,$$

with no age-by-group product term on the log scale. The figure plots a Poisson-log model fitted to one such sample, on the count scale and on the log scale. The count scenario of the diagnostic simulation uses the same structure with a slightly lower intercept (2.20) and group effect (0.45).

The motivating 2×2 figure is fully deterministic: it evaluates four cell probabilities, without sampling. They are generated additively on the chance-corrected logit scale with $c = .50$,

$$p = .50 + .50\,\text{logistic}(\eta), \qquad \eta = -1.00 + 2.10\,\text{condition} + 1.70\,\text{group},$$

again with no condition-by-group product term. The resulting cell probabilities are given in Table 6. Because a saturated 2×2 design has exactly one product-term coefficient per scale, those same four probabilities imply the three different interaction coefficients reported in Table 7: exactly zero on the generating scale, positive on the standard logit scale, and negative on the raw probability scale.

Table 6
Cell probabilities of the motivating 2 x 2 figure. These four values are identical in all three panels of that figure.

Group	Condition	Linear predictor	Expected probability
Group 0	0	-1.00	0.634
Group 0	1	1.10	0.875
Group 1	0	0.70	0.834
Group 1	1	2.80	0.971

Table 7
Product-term coefficient implied by the cell probabilities in the previous table, on each of the three scales shown in the motivating figure. Only the scale on which the difference between differences is computed changes.

Scale	Implied product term
Linear (probability)	-0.103
Standard logit	0.513
Chance-corrected logit	0.000

Simulation 1: forced-choice accuracy with a chance floor

For each participant i, age is drawn on the range 6 to 10 and centered at 8, group is a two-level factor, and the number of correct responses out of $k = 20$ trials is binomial with

$$p_i = .50 + .50\,\text{logistic}(\eta_i), \qquad \eta_i = \beta_0 + 0.60\,\text{age}_c - 0.90\,\text{group} + 0\,(\text{age}_c \times \text{group}).$$

The generating scale is the chance-corrected logit with a fixed floor of .50, and the age-by-group product term is exactly zero on that scale. Three scenarios move the same structure along the accuracy range through the intercept: lower performance ($\beta_0 = -0.80$), middle performance ($\beta_0 = 0.00$), and higher performance ($\beta_0 = 0.80$). Each replication samples $N = 250$ participants.

Four candidate models are fitted to each replication with the same `age_c * group` interaction formula: (1) a Gaussian identity model on the accuracy proportion; (2) a standard binomial logit on successes out of trials; (3) a standard binomial probit; and (4) a chance-corrected binomial logit fitted by direct maximum likelihood on the binomial likelihood with mean $p = c + (1 - c)\,\text{logistic}(\eta)$ and $c = .50$ fixed by design. For the chance-corrected fit, starting values come from a standard binomial fit, optimization uses BFGS with a numerically computed Hessian, and Wald p-values are derived from the inverse Hessian. Replications in which optimization or the Hessian inversion fails are recorded as failed fits and summarized over successful fits only. The chance-corrected model can fail when observed performance sits close to the floor, where the likelihood carries little information about above-chance changes; the main paper treats this as information about the design rather than as a numerical nuisance.

Table 8

Product-term rejection rates by forced-choice performance scenario and fitted model. The matched chance-corrected model shows nominal rejection; alternative scales show pseudo-interaction detection. Fits = converged replications; Reject = rejected product-term tests; CI = Wilson 95%.

Scenario	Model	Rate type	Fits	Reject	Rejection rate [95% CI]
Lower performance	Gaussian identity	Pseudo-interaction detection rate	300	137	45.7% [40.1, 51.3]
Lower performance	Standard binomial logit	Pseudo-interaction detection rate	300	191	63.7% [58.1, 68.9]
Lower performance	Standard binomial probit	Pseudo-interaction detection rate	300	183	61.0% [55.4, 66.3]

Scenario	Model	Rate type	Fits	Reject	Rejection rate [95% CI]
Lower performance	Chance-corrected binomial logit	Nominal rejection rate	179	13	7.3% [4.3, 12.0]
Middle performance	Gaussian identity	Pseudo-interaction detection rate	300	40	13.3% [9.9, 17.6]
Middle performance	Standard binomial logit	Pseudo-interaction detection rate	300	169	56.3% [50.7, 61.8]
Middle performance	Standard binomial probit	Pseudo-interaction detection rate	300	148	49.3% [43.7, 55.0]
Middle performance	Chance-corrected binomial logit	Nominal rejection rate	300	18	6.0% [3.8, 9.3]
Higher performance	Gaussian identity	Pseudo-interaction detection rate	300	36	12.0% [8.8, 16.2]
Higher performance	Standard binomial logit	Pseudo-interaction detection rate	300	103	34.3% [29.2, 39.9]
Higher performance	Standard binomial probit	Pseudo-interaction detection rate	300	59	19.7% [15.6, 24.5]
Higher performance	Chance-corrected binomial logit	Nominal rejection rate	300	14	4.7% [2.8, 7.7]

Because the chance-corrected model is the only one that can fail to converge, the rates in Table 8 are not all computed on the same replications: in the lower-performance scenario the three alternative models use all 300 replications while the matched model uses the 179 that converged. Table 9 repeats all four models on the replications in which *every* model returned a usable interaction test, so that within each scenario the four rates describe identical datasets. The middle- and higher-performance scenarios are unaffected, because all four models converged everywhere there. In the lower-performance scenario the rates shift slightly, and the ordering of the four models is unchanged: the standard logit remains highest, followed by the standard probit, the Gaussian identity, and the matched chance-corrected model near nominal alpha. The selective convergence therefore does not create the pattern reported in the main paper.

Table 9
Product-term rejection rates for the forced-choice simulation, restricted to the replications in which all four models converged. Common reps = replications retained in the scenario, identical for the four models by construction; Reject is computed within that subset. CI = Wilson 95%.

Scenario	Model	Rate type	Common reps	Reject	Rejection rate [95% CI]
Lower performance	Gaussian identity	Pseudo-interaction detection rate	179	79	44.1% [37.1, 51.5]
Lower performance	Standard binomial logit	Pseudo-interaction detection rate	179	111	62.0% [54.7, 68.8]
Lower performance	Standard binomial probit	Pseudo-interaction detection rate	179	108	60.3% [53.0, 67.2]
Lower performance	Chance-corrected binomial logit	Nominal rejection rate	179	13	7.3% [4.3, 12.0]
Middle performance	Gaussian identity	Pseudo-interaction detection rate	300	40	13.3% [9.9, 17.6]
Middle performance	Standard binomial logit	Pseudo-interaction detection rate	300	169	56.3% [50.7, 61.8]
Middle performance	Standard binomial probit	Pseudo-interaction detection rate	300	148	49.3% [43.7, 55.0]
Middle performance	Chance-corrected binomial logit	Nominal rejection rate	300	18	6.0% [3.8, 9.3]
Higher performance	Gaussian identity	Pseudo-interaction detection rate	300	36	12.0% [8.8, 16.2]

Scenario	Model	Rate type	Common reps	Reject	Rejection rate [95% CI]
Higher performance	Standard binomial logit	Pseudo-interaction detection rate	300	103	34.3% [29.2, 39.9]
Higher performance	Standard binomial probit	Pseudo-interaction detection rate	300	59	19.7% [15.6, 24.5]
Higher performance	Chance-corrected binomial logit	Nominal rejection rate	300	14	4.7% [2.8, 7.7]

Simulation 2: within-family link choice

Repeated binary trials are generated in a two-group, two-condition design with $n = 700$ participants, $k = 15$ trials per participant-condition cell, and a subject random intercept. The reference fixed effects on the probit scale are intercept 1.50, group -1.00, condition -1.00, and product term 0. When the generating link is logit, all fixed effects are multiplied by 1.65, so that the logit and probit data-generating processes imply similar cell probabilities while preserving a zero product term on the generating link scale. The random-intercept standard deviation is set from a latent-scale ICC of .30, using the latent residual variance of the generating link ($\pi^2/3$ for logit, 1 for probit).

Generating and fitted links are fully crossed (logit and probit in both roles), giving two matched-link and two wrong-link cells. All models are random-intercept GLMMs fitted with `glmmTMB`, using the same `group * condition` fixed-effect formula. The separate fitted-curve illustration uses a grouped-binomial dataset with 1,000 observational units and 60 trials per unit, with one `cbind(correct, incorrect)` response per unit; it is shown in Section S5, Figure 1.

Table 10

Product-term rejection rates for the within-family link simulation. Matched links show nominal rejection; wrong links show pseudo-interaction detection. The median coefficient and SE are on the fitted link scale. CI = Wilson 95%.

Generating	Fitted	Status	Rate type	Fits	Reject	Rejection rate [95% CI]	Median beta	Median SE
logit	logit	Matched link	Nominal rejection rate	300	16	5.3% [3.3, 8.5]	-0.001	0.075
logit	probit	Wrong link	Pseudo-interaction detection rate	300	45	15.0% [11.4, 19.5]	-0.041	0.042
probit	logit	Wrong link	Pseudo-interaction detection rate	300	41	13.7% [10.2, 18.0]	0.071	0.076
probit	probit	Matched link	Nominal rejection rate	300	15	5.0% [3.1, 8.1]	-0.003	0.043

Simulation 3: sum scores

Each replication samples $N = 600$ participants. A latent trait is generated as

$$\theta_i = 0.85\,x_i - 0.90\,\text{group}_i + 0\,(x_i \times \text{group}_i) + \varepsilon_i, \qquad \varepsilon_i \sim N(0,1),$$

so the x-by-group product term is exactly zero on the latent generating scale. The latent trait is mapped to $J = 9$ ordinal items, each scored 0 to 3, through a graded logistic threshold model: item j has three ordered thresholds, and the probability of reaching category k or higher is logistic$(\theta_i - \tau_{jk})$. Thresholds are deterministic, with base values $(-1, 0, 1)$, evenly spaced item offsets from -0.45 to 0.45 across the nine items, and a scenario-level shift: middle of scale (shift 0.00), near floor (shift $+1.60$, harder items compress scores toward the lower bound), and near ceiling (shift -1.60). The observed outcome is the item total, bounded between 0 and 27, discrete, and differentially compressed across the score range.

Three analyses are fitted with the same `x * group` formula, to separate the relevant scales: (1) an observed sum-score identity model, `lm()` on the manifest total; (2) a Gaussian-probit bounded-score model, a Gaussian GLM with a probit link fitted to the continuity-corrected score proportion $(\text{sum} + 0.5)/(\text{max} + 1)$, included as a sensitivity model on the manifest total rather than a measurement model; and (3) a latent-scale benchmark, `lm()` on the true θ, available only because θ is known in the simulation.

Table 11

Product-term rejection rates by sum-score scenario and analysis scale. The latent generating-scale benchmark shows nominal rejection; observed-score models show pseudo-interaction detection. The latent benchmark is available only because theta is known. CI = Wilson 95%.

Scenario	Model	Rate type	Fits	Reject	Rejection rate [95% CI]
Lower range	Latent generating scale	Nominal rejection rate	300	16	5.3% [3.3, 8.5]
Lower range	Observed sum-score identity	Pseudo-interaction detection rate	300	211	70.3% [64.9, 75.2]
Lower range	Gaussian-probit bounded score	Pseudo-interaction detection rate	300	30	10.0% [7.1, 13.9]
Middle range	Latent generating scale	Nominal rejection rate	300	18	6.0% [3.8, 9.3]
Middle range	Observed sum-score identity	Pseudo-interaction detection rate	300	25	8.3% [5.7, 12.0]
Middle range	Gaussian-probit bounded score	Pseudo-interaction detection rate	300	10	3.3% [1.8, 6.0]
Upper range	Latent generating scale	Nominal rejection rate	300	16	5.3% [3.3, 8.5]
Upper range	Observed sum-score identity	Pseudo-interaction detection rate	300	112	37.3% [32.1, 42.9]
Upper range	Gaussian-probit bounded score	Pseudo-interaction detection rate	300	15	5.0% [3.1, 8.1]

Diagnostic simulation: data-generating processes

The main paper introduces the five diagnostic scenarios in prose. Their formal generating models are given here. In every scenario the product term is exactly zero on the generating link scale, and the fitted model keeps the outcome family and the interaction formula while changing only the link.

D1, count errors (Poisson log, fitted with identity). For $N = 250$ children with age uniform on $[6, 10]$ and a balanced group factor, errors are Poisson with

$$\log \mu_i = 2.20 - 0.55\,(\text{age}_i - 6) + 0.45\,\text{group}_i .$$

D2, chance-floor accuracy (chance-corrected logit, fitted with standard logit). The lower-performance scenario of Simulation 1: $k = 20$ trials per participant and

$$p_i = .50 + .50\,\text{logistic}(\eta_i), \qquad \eta_i = -0.80 + 0.60\,\text{age}_{c,i} - 0.90\,\text{group}_i .$$

D3, binary repeated trials (probit, fitted with logit). A two-group, two-condition design with 15 trials per participant-condition cell, a subject random intercept, and a latent ICC of .30:

$$\Pr(y_{ij} = 1) = \Phi\big(1.50 - 1.00\,\text{group}_i - 1.00\,\text{condition}_j + u_i\big) .$$

D4, binary with a continuous predictor (probit, fitted with logit). The same structure as D3, with the two-level condition replaced by a continuous predictor $x \sim \text{Uniform}(0, 1)$, so that the contrast from $x = 0$ to $x = 1$ has the same probit-scale size as the condition contrast in D3. This scenario exists because residual checks over a focal predictor are better identified when the predictor has many unique values.

D5, Gamma mean response time (log, fitted with inverse). Response times are Gamma with shape 8 and

$$\log \mu_i = \log(400) + 0.25\, x_i + 0.25\, \text{group}_i,$$

so the baseline mean is 400 ms at $x = 0$ in the reference group.

Table 12 reports the expected values these parameters imply, which is what the misspecified link has to reproduce with a product term.

Table 12

Expected values implied by the five diagnostic data-generating processes, at low, middle, and high values of the focal predictor. Values for the two binary scenarios are conditional on a subject random intercept of zero. Units differ by scenario and are given in the outcome column.

Scenario	Predictor	Value	Group	Linear predictor	Outcome	Expected
Count errors	age	6.0	Group 0	2.20	expected count	9.025
Count errors	age	8.0	Group 0	1.10	expected count	3.004
Count errors	age	10.0	Group 0	0.00	expected count	1.000
Count errors	age	6.0	Group 1	2.65	expected count	14.154
Count errors	age	8.0	Group 1	1.55	expected count	4.711
Count errors	age	10.0	Group 1	0.45	expected count	1.568
Chance-floor accuracy	age	6.0	Group 0	-2.00	expected accuracy	0.560
Chance-floor accuracy	age	8.0	Group 0	-0.80	expected accuracy	0.655

Scenario	Predictor	Value	Group	Linear predictor	Outcome	Expected
Chance-floor accuracy	age	10.0	Group 0	0.40	expected accuracy	0.799
Chance-floor accuracy	age	6.0	Group 1	-2.90	expected accuracy	0.526
Chance-floor accuracy	age	8.0	Group 1	-1.70	expected accuracy	0.577
Chance-floor accuracy	age	10.0	Group 1	-0.50	expected accuracy	0.689
Binary repeated trials	condition	0.0	Group 0	1.50	expected probability (u = 0)	0.933
Binary repeated trials	condition	1.0	Group 0	0.50	expected probability (u = 0)	0.691
Binary repeated trials	condition	0.0	Group 1	0.50	expected probability (u = 0)	0.691
Binary repeated trials	condition	1.0	Group 1	-0.50	expected probability (u = 0)	0.309
Binary continuous predictor	x	0.0	Group 0	1.50	expected probability (u = 0)	0.933
Binary continuous predictor	x	1.0	Group 0	0.50	expected probability (u = 0)	0.691
Binary continuous predictor	x	0.0	Group 1	0.50	expected probability (u = 0)	0.691
Binary continuous predictor	x	1.0	Group 1	-0.50	expected probability (u = 0)	0.309
Gamma mean response time	x	-2.0	Group 0	5.49	expected mean (ms)	242.612
Gamma mean response time	x	0.0	Group 0	5.99	expected mean (ms)	400.000
Gamma mean response time	x	2.0	Group 0	6.49	expected mean (ms)	659.489
Gamma mean response time	x	-2.0	Group 1	5.74	expected mean (ms)	311.520
Gamma mean response time	x	0.0	Group 1	6.24	expected mean (ms)	513.610
Gamma mean response time	x	2.0	Group 1	6.74	expected mean (ms)	846.800

Diagnostic simulation: estimation and checks

The diagnostic simulation ties each of the five link-misspecification scenarios defined above to a misspecified fitted link, holding the outcome family and the interaction formula fixed. In every replication the interaction model is fitted under the wrong link, and the primary diagnostics are applied to that same fitted model, because that is the model an analyst would inspect after finding the interaction. The same diagnostics are also applied to the corresponding correct-link interaction model as a calibration check. The technical specification of the checks is as follows.

Simulation-based residuals (Dunn & Smyth, 1996; Hartig, 2024) were computed with DHARMa, simulating 250 responses from the fitted model per replication. Four prespecified checks were applied: residual uniformity (`testUniformity`), dispersion (`testDispersion`), residual quantile patterns over fitted values (`testQuantiles`), and residual structure over the focal predictor. The last check is scenario-aware: `testQuantiles` over the predictor is used when the predictor has at least eight unique values; for the 2×2 repeated-trials scenario, residuals are instead compared across the group-by-condition design cells using a Kruskal-Wallis test on the DHARMa scaled residuals. Table 19 reports this design-cell check for D3. Cells are marked "–" only when a check is not applicable or returns no usable replications in the saved diagnostic summary.

The Pregibon-style added-term link check (Pregibon, 1980) extracts the fitted linear predictor, adds its square to the model formula, and tests the significance of the added term while retaining the family, link, base formula, and random-effects structure of the model being diagnosed.

The chance-corrected fits in D2 are the only ones estimated outside a standard model-fitting function, and they use a more defensive procedure than Simulation 1, because here the target-link model is one of the two candidates in the AIC comparison and a failed fit would silently remove a replication from that comparison. The model is fitted by direct maximum likelihood with $c = .50$ fixed by design. Five starting values are tried: a vector of zeros, three derived from the corresponding standard-logit fit, and one from a linear fit on the empirical above-chance proportions, which is the scale the likelihood actually works on. Each start is optimized with both BFGS and Nelder–Mead, the solution with the highest likelihood is retained, and it is then polished with a tighter BFGS run. Because `optim()` reports only that a stopping rule was met, and that rule is easily met far from the maximum on the flat region near the chance floor, the retained solution is additionally checked against the score and the curvature of the log-likelihood; the same Hessian supplies the variance-covariance matrix used for the Wald test. Replications that fail these checks are recorded as failed fits.

The same-formula AIC comparison contrasts the target-link and the misspecified-link interaction models, holding the response data, the likelihood family, and the structural formula fixed, so only the mean-link mapping varies. No inconclusive ΔAIC category is used, so the two AIC rows in Table 19 are complementary whenever both AIC values are available.

S5. Supplementary tables and figures

The tables in this section report quantities at a finer grain than the main paper, or quantities the main paper states only in prose or in a figure. The headline rejection-rate summaries are reported in Section S4, next to each data-generating process.

Implied values, Simulation 1

The main paper describes the three forced-choice scenarios by the accuracy range they occupy. Table 13 gives the exact expected values behind those descriptions, and Table 14 gives the contrasts derived from them. The last row of the contrast table is the quantity the simulation is about: the age-by-group product term is exactly zero on the generating chance-corrected logit scale, while the group difference in expected accuracy still changes with age, by an amount that grows as the scenario moves toward the chance floor.

Table 13

Expected accuracy implied by the three forced-choice scenarios, at the youngest, middle, and oldest age. Correct responses are out of k = 20 trials.

Scenario	Age	Group	Linear predictor	Expected accuracy	Correct out of 20
Lower performance	6	Group 0	-2.00	0.560	11.19
Lower performance	8	Group 0	-0.80	0.655	13.10
Lower performance	10	Group 0	0.40	0.799	15.99
Lower performance	6	Group 1	-2.90	0.526	10.52
Lower performance	8	Group 1	-1.70	0.577	11.54
Lower performance	10	Group 1	-0.50	0.689	13.78
Middle performance	6	Group 0	-1.20	0.616	12.31
Middle performance	8	Group 0	0.00	0.750	15.00
Middle performance	10	Group 0	1.20	0.884	17.69
Middle performance	6	Group 1	-2.10	0.555	11.09
Middle performance	8	Group 1	-0.90	0.645	12.89
Middle performance	10	Group 1	0.30	0.787	15.74
Higher performance	6	Group 0	-0.40	0.701	14.01
Higher performance	8	Group 0	0.80	0.845	16.90
Higher performance	10	Group 0	2.00	0.940	18.81
Higher performance	6	Group 1	-1.30	0.607	12.14
Higher performance	8	Group 1	-0.10	0.738	14.75
Higher performance	10	Group 1	1.10	0.875	17.50

Table 14

Contrasts derived from the forced-choice scenarios. Probability points and correct-response units are on the observed accuracy scale; the link-scale column is on the generating chance-corrected logit scale.

Scenario	Contrast	Probability points	Correct out of 20	Link scale
Lower performance	Group difference at youngest age: Group 1 minus Group 0	-0.034	-0.67	–
Lower performance	Group difference at oldest age: Group 1 minus Group 0	-0.111	-2.21	–
Lower performance	Age-related change in Group 0: oldest minus youngest	0.240	4.79	–
Lower performance	Age-related change in Group 1: oldest minus youngest	0.163	3.25	–
Lower performance	Change in group difference from youngest to oldest age	-0.077	-1.54	–
Lower performance	Generating link-scale age-by-group product term	–	–	0.000
Middle performance	Group difference at youngest age: Group 1 minus Group 0	-0.061	-1.22	–
Middle performance	Group difference at oldest age: Group 1 minus Group 0	-0.097	-1.94	–
Middle performance	Age-related change in Group 0: oldest minus youngest	0.269	5.37	–
Middle performance	Age-related change in Group 1: oldest minus youngest	0.233	4.65	–
Middle performance	Change in group difference from youngest to oldest age	-0.036	-0.72	–

Scenario	Contrast	Probability points	Correct out of 20	Link scale
Middle performance	Generating link-scale age-by-group product term	–	–	0.000
Higher performance	Group difference at youngest age: Group 1 minus Group 0	-0.094	-1.87	–
Higher performance	Group difference at oldest age: Group 1 minus Group 0	-0.065	-1.31	–
Higher performance	Age-related change in Group 0: oldest minus youngest	0.240	4.79	–
Higher performance	Age-related change in Group 1: oldest minus youngest	0.268	5.36	–
Higher performance	Change in group difference from youngest to oldest age	0.028	0.57	–
Higher performance	Generating link-scale age-by-group product term	–	–	0.000

Deterministic link-scale decomposition, Simulation 2

The nominal rejection and pseudo-interaction detection rates for Simulation 2 are reported in Section S4. Table 15 instead reports the deterministic quantities behind them: for each generating-by-fitted link combination, the product-term contrast induced by evaluating the generating cell probabilities on the fitted link scale, and the corresponding response-scale difference-in-differences. Under a matched link the induced product term is zero by construction; under a wrong link it is a fixed nonzero value. The wrong-link rejection rate reflects power to detect that fitted-scale product term.

Table 15

Deterministic link-scale decomposition for Simulation 2. The induced product term is evaluated on the fitted link scale; the response-scale difference-in-differences is on the probability scale.

Generating link	Fitted link	Match	Induced product (fitted scale)	Response-scale diff-in-diff
logit	logit	Matched link	0.000	-0.164
logit	probit	Wrong link	-0.112	-0.164
probit	logit	Wrong link	0.216	-0.141
probit	probit	Matched link	0.000	-0.141

Implied values, Simulation 3

Table 16 gives the expected sum scores implied by the three sum-score scenarios, and Table 17 the contrasts derived from them. The generating latent-scale product term is zero in all three, yet the change in the group difference from low to high x, expressed in raw score points, is close to zero only in the middle of the scale and grows as the expected totals approach a bound. This is the compression the main paper describes, stated as a number rather than as a curve.

Table 16

Expected sum scores implied by the three sum-score scenarios, at low, middle, and high values of the predictor. Totals run from 0 to 27.

Scenario	x	Group	Latent mean	Expected sum score	Percent of max
Lower range	-1	Group 0	-0.85	3.52	13.1%
Lower range	0	Group 0	0.00	6.20	23.0%
Lower range	1	Group 0	0.85	9.82	36.4%
Lower range	-1	Group 1	-1.75	1.75	6.5%

Scenario	x	Group	Latent mean	Expected sum score	Percent of max
Lower range	0	Group 1	-0.90	3.40	12.6%
Lower range	1	Group 1	-0.05	6.01	22.3%
Middle range	-1	Group 0	-0.85	9.35	34.6%
Middle range	0	Group 0	0.00	13.50	50.0%
Middle range	1	Group 0	0.85	17.65	65.4%
Middle range	-1	Group 1	-1.75	5.65	20.9%
Middle range	0	Group 1	-0.90	9.12	33.8%
Middle range	1	Group 1	-0.05	13.25	49.1%
Upper range	-1	Group 0	-0.85	17.18	63.6%
Upper range	0	Group 0	0.00	20.80	77.0%
Upper range	1	Group 0	0.85	23.48	86.9%
Upper range	-1	Group 1	-1.75	12.75	47.2%
Upper range	0	Group 1	-0.90	16.94	62.8%
Upper range	1	Group 1	-0.05	20.61	76.3%

Table 17

Contrasts derived from the sum-score scenarios. Sum-score units and percent of maximum are on the observed total; the latent-scale column is on the generating scale.

Scenario	Contrast	Sum-score units	Percent of max	Latent scale
Lower range	Group difference at low x: Group 1 minus Group 0	-1.774	-6.6%	–
Lower range	Group difference at high x: Group 1 minus Group 0	-3.807	-14.1%	–

Scenario	Contrast	Sum-score units	Percent of max	Latent scale
Lower range	x-related change in Group 0: high x minus low x	6.295	23.3%	–
Lower range	x-related change in Group 1: high x minus low x	4.262	15.8%	–
Lower range	Change in group difference from low x to high x	-2.034	-7.5%	–
Lower range	Generating latent-scale x-by-group product term	–	–	0.000
Middle range	Group difference at low x: Group 1 minus Group 0	-3.702	-13.7%	–
Middle range	Group difference at high x: Group 1 minus Group 0	-4.398	-16.3%	–
Middle range	x-related change in Group 0: high x minus low x	8.294	30.7%	–
Middle range	x-related change in Group 1: high x minus low x	7.598	28.1%	–
Middle range	Change in group difference from low x to high x	-0.696	-2.6%	–
Middle range	Generating latent-scale x-by-group product term	–	–	0.000
Upper range	Group difference at low x: Group 1 minus Group 0	-4.432	-16.4%	–
Upper range	Group difference at high x: Group 1 minus Group 0	-2.863	-10.6%	–
Upper range	x-related change in Group 0: high x minus low x	6.295	23.3%	–
Upper range	x-related change in Group 1: high x minus low x	7.865	29.1%	–
Upper range	Change in group difference from low x to high x	1.570	5.8%	–
Upper range	Generating latent-scale x-by-group product term	–	–	0.000

Logit and probit fitted curves

Section S4 notes that the within-family comparison is also illustrated with a separate grouped-binomial dataset of 1,000 observational units and 60 trials per unit. Figure 1 shows the two fitted curves, and Table 18 the corresponding coefficients and AIC values.

The illustration makes concrete why the two links are usually treated as interchangeable and why that is not sufficient for an interaction claim. Panel A shows fitted probabilities that are hard to tell apart by eye. Panel B plots their difference, which never exceeds about two probability points but is systematic rather than noisy: it changes sign across the predictor range, so the logit fit sits above the probit fit in some regions and below it in others. A difference between differences computed on one scale therefore cannot equal the same contrast computed on the other, which is the deterministic quantity tabulated in Table 15. The coefficients in Table 18 are on each model's own link scale and are not comparable across rows; the AIC values are comparable, and they differ by far more than the fitted probabilities do.

Figure 1

Logit and probit models fitted to the same grouped-binomial dataset.

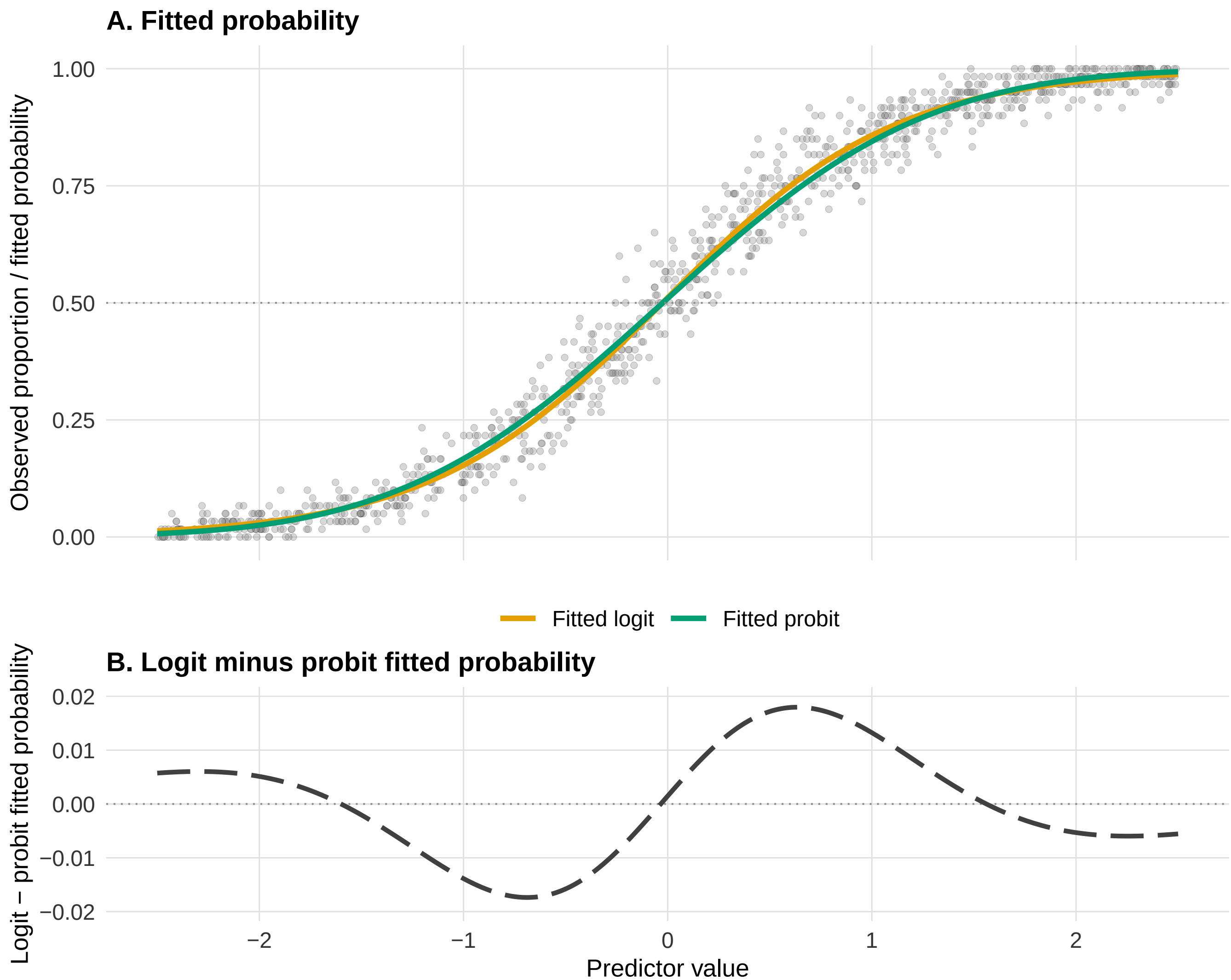


Table 18

Coefficients and AIC for the logit and probit models fitted to the grouped-binomial illustration dataset. Intercept and slope are on each model's own link scale, so they are not directly comparable across rows.

Model	Intercept	Slope	AIC
logit	0.045	1.753	4315.6
probit	0.024	0.990	4380.4

Diagnostic detection rates and AIC strength

The main paper reports the DHARMa detection rate as a min-to-max range across checks. Table 19 gives the individual checks, the within-replication alignment between pseudo-interactions and applicable DHARMa checks, and the strength of the same-formula AIC comparison.

Table 19

Diagnostic detection rates and AIC strength across five scenarios. D1 count errors (Poisson log to identity); D2 chance-floor accuracy (chance-corrected to standard logit); D3 binary 2x2 GLMM (probit to logit); D4 binary continuous-predictor GLMM (probit to logit); D5 Gamma mean response time (log to inverse).

Check	D1	D2	D3	D4	D5
Pseudo-interaction (wrong link)	269/281 (95.7%)	177/300 (59.0%)	34/300 (11.3%)	21/300 (7.0%)	92/300 (30.7%)
DHARMa: uniformity	0.0%	0.0%	0.7%	1.7%	0.0%
DHARMa: dispersion	24.9%	0.0%	0.0%	0.0%	0.3%
DHARMa: quantiles vs fitted	60.5%	1.7%	0.7%	2.3%	2.0%
DHARMa: quantiles vs predictor	88.3%	5.0%	–	1.7%	5.3%

Check	D1	D2	D3	D4	D5
DHARMa: across design cells	–	–	0.0%	–	–
Pseudo-interactions missed by all applicable DHARMa checks	25/269 (9.3%)	166/177 (93.8%)	33/34 (97.1%)	21/21 (100.0%)	87/92 (94.6%)
Pregibon link check	100.0%	29.0%	100.0%	100.0%	35.3%
AIC favors target link	98.6%	78.3%	93.7%	94.0%	78.0%
AIC favors misspecified link	1.4%	21.7%	6.3%	6.0%	22.0%
Median ΔAIC	22.5 (n = 281)	2.2 (n = 300)	6.1 (n = 300)	4.9 (n = 300)	2.6 (n = 300)
|ΔAIC| < 2	5/281 (1.8%)	119/300 (39.7%)	40/300 (13.3%)	54/300 (18.0%)	106/300 (35.3%)

Pseudo-interaction denominators are replications with an available interaction test. “Missed” means that none of the applicable prespecified DHARMa checks had $p < .05$; pseudo-interactions for which all five DHARMa p-values were missing are excluded from this conditional denominator. $\Delta\text{AIC} = \text{AIC(wrong link)} - \text{AIC(target link)}$, so positive values favor the target link and negative values favor the wrong link. The sample size shown with the median and the denominator for $|\Delta\text{AIC}| < 2$ are the finite AIC comparisons. The Pregibon row reaches 100% in D1, D3, and D4, but only the D1 value is evidence about the link: as the calibration section below shows, in D3 and D4 the same check also flags every replication when the link is correct.

Distribution of the AIC difference

The main paper argues that the proportion of replications in which AIC favours the target link is not informative on its own, because winning by less than one AIC unit and winning by fifty describe different situations. Table 20 therefore gives the distribution of ΔAIC rather than only its median. The contrast between the count scenario and the chance-floor scenario is the clearest: both favour the target link in most replications, but the count scenario separates the two links by tens of AIC units across the whole distribution, whereas in the chance-floor scenario the lower decile is negative, meaning that in a non-trivial share of replications AIC actively preferred the misspecified link, and roughly two fifths of the comparisons fall within two units of a tie.

Table 20

Distribution of the AIC difference across replications, by diagnostic scenario. Delta AIC = AIC(wrong link) - AIC(target link), so positive values favor the target link. n = replications with both AIC values available.

Scenario	n	10th	25th	Median	75th	90th	Within 2 units	Favors target
Count errors	281	10.0	16.0	22.5	29.9	35.5	1.8%	98.6%
Chance-floor accuracy	300	-1.2	0.4	2.2	4.1	6.8	39.7%	78.3%
Binary repeated trials	300	0.9	3.4	6.1	8.0	10.3	13.3%	93.7%
Binary continuous predictor	300	0.7	2.8	4.9	6.8	8.8	18.0%	94.0%
Gamma mean response time	300	-1.2	0.6	2.6	4.7	6.7	35.3%	78.0%

Calibration under correctly specified links

The same DHARMa and Pregibon-style checks were applied to the corresponding correct-link interaction model in each replication. These baseline flagging rates provide a reference for interpreting detection under the wrong link. They reuse the same simulated datasets and scenarios rather than defining a new simulation.

Two patterns in Table 21 are worth reading before the wrong-link column. The four DHARMa checks are well calibrated throughout: no scenario exceeds a correct-link flagging rate of about six percent, so their wrong-link rates can be read as detection. The Pregibon-style check is well calibrated in the same sense in the count, chance-floor, and Gamma scenarios, but not in the two random-intercept GLMM scenarios, where it flags every correct-link replication. In a mixed model the added $\hat{\eta}^2$ term appears to absorb structure that is present whether or not the link is correct, so in those two scenarios the check is degenerate rather than powerful, and its wrong-link rate of 100% carries no information about link adequacy. A diagnostic has to be calibrated under the intended model before its behaviour under an alternative can be interpreted, and this is the one place in the present set of checks where that condition fails.

Table 21

Diagnostic calibration under correctly specified and misspecified links. Each cell gives the flagging rate with its Wilson 95% confidence interval and the number of successful diagnostic fits over attempts.

Scenario	Diagnostic	Correct-link flagging	Wrong-link flagging
Count errors	DHARMa uniformity	0.0% [0.0, 1.3] (300/300)	0.0% [0.0, 1.3] (281/300)
Count errors	DHARMa dispersion	4.7% [2.8, 7.7] (300/300)	24.9% [20.2, 30.3] (281/300)
Count errors	DHARMa residual quantiles over fitted values	1.7% [0.7, 3.8] (300/300)	60.5% [54.7, 66.0] (281/300)
Count errors	DHARMa residual quantiles over focal predictor	2.7% [1.4, 5.2] (300/300)	88.3% [84.0, 91.5] (281/300)
Count errors	Pregibon-style added-term link check	5.3% [3.3, 8.5] (300/300)	100.0% [20.7, 100.0] (1/300)
Chance-floor accuracy	DHARMa uniformity	0.0% [0.0, 1.3] (300/300)	0.0% [0.0, 1.3] (300/300)
Chance-floor accuracy	DHARMa dispersion	0.7% [0.2, 2.4] (300/300)	0.0% [0.0, 1.3] (300/300)
Chance-floor accuracy	DHARMa residual quantiles over fitted values	1.3% [0.5, 3.4] (300/300)	1.7% [0.7, 3.8] (300/300)
Chance-floor accuracy	DHARMa residual quantiles over focal predictor	1.7% [0.7, 3.8] (300/300)	5.0% [3.1, 8.1] (300/300)
Chance-floor accuracy	Pregibon-style added-term link check	5.7% [3.6, 8.9] (300/300)	29.0% [24.2, 34.4] (300/300)

Scenario	Diagnostic	Correct-link flagging	Wrong-link flagging
Binary repeated trials	DHARMa uniformity	0.3% [0.1, 1.9] (300/300)	0.7% [0.2, 2.4] (300/300)
Binary repeated trials	DHARMa dispersion	0.0% [0.0, 1.3] (300/300)	0.0% [0.0, 1.3] (300/300)
Binary repeated trials	DHARMa residual quantiles over fitted values	0.0% [0.0, 1.3] (300/300)	0.7% [0.2, 2.4] (300/300)
Binary repeated trials	DHARMa residual distribution across design cells	0.0% [0.0, 1.3] (300/300)	0.0% [0.0, 1.3] (300/300)
Binary repeated trials	Pregibon-style added-term link check	100.0% [98.7, 100.0] (300/300)	100.0% [98.7, 100.0] (300/300)
Binary continuous predictor	DHARMa uniformity	2.0% [0.9, 4.3] (300/300)	1.7% [0.7, 3.8] (300/300)
Binary continuous predictor	DHARMa dispersion	0.0% [0.0, 1.3] (300/300)	0.0% [0.0, 1.3] (300/300)
Binary continuous predictor	DHARMa residual quantiles over fitted values	1.3% [0.5, 3.4] (300/300)	2.3% [1.1, 4.7] (300/300)
Binary continuous predictor	DHARMa residual quantiles over focal predictor	2.3% [1.1, 4.7] (300/300)	1.7% [0.7, 3.8] (300/300)
Binary continuous predictor	Pregibon-style added-term link check	100.0% [98.7, 100.0] (300/300)	100.0% [98.7, 100.0] (300/300)
Gamma mean response time	DHARMa uniformity	0.0% [0.0, 1.3] (300/300)	0.0% [0.0, 1.3] (300/300)
Gamma mean response time	DHARMa dispersion	0.3% [0.1, 1.9] (300/300)	0.3% [0.1, 1.9] (300/300)
Gamma mean response time	DHARMa residual quantiles over fitted values	0.7% [0.2, 2.4] (300/300)	2.0% [0.9, 4.3] (300/300)
Gamma mean response time	DHARMa residual quantiles over focal predictor	1.3% [0.5, 3.4] (300/300)	5.3% [3.3, 8.5] (300/300)

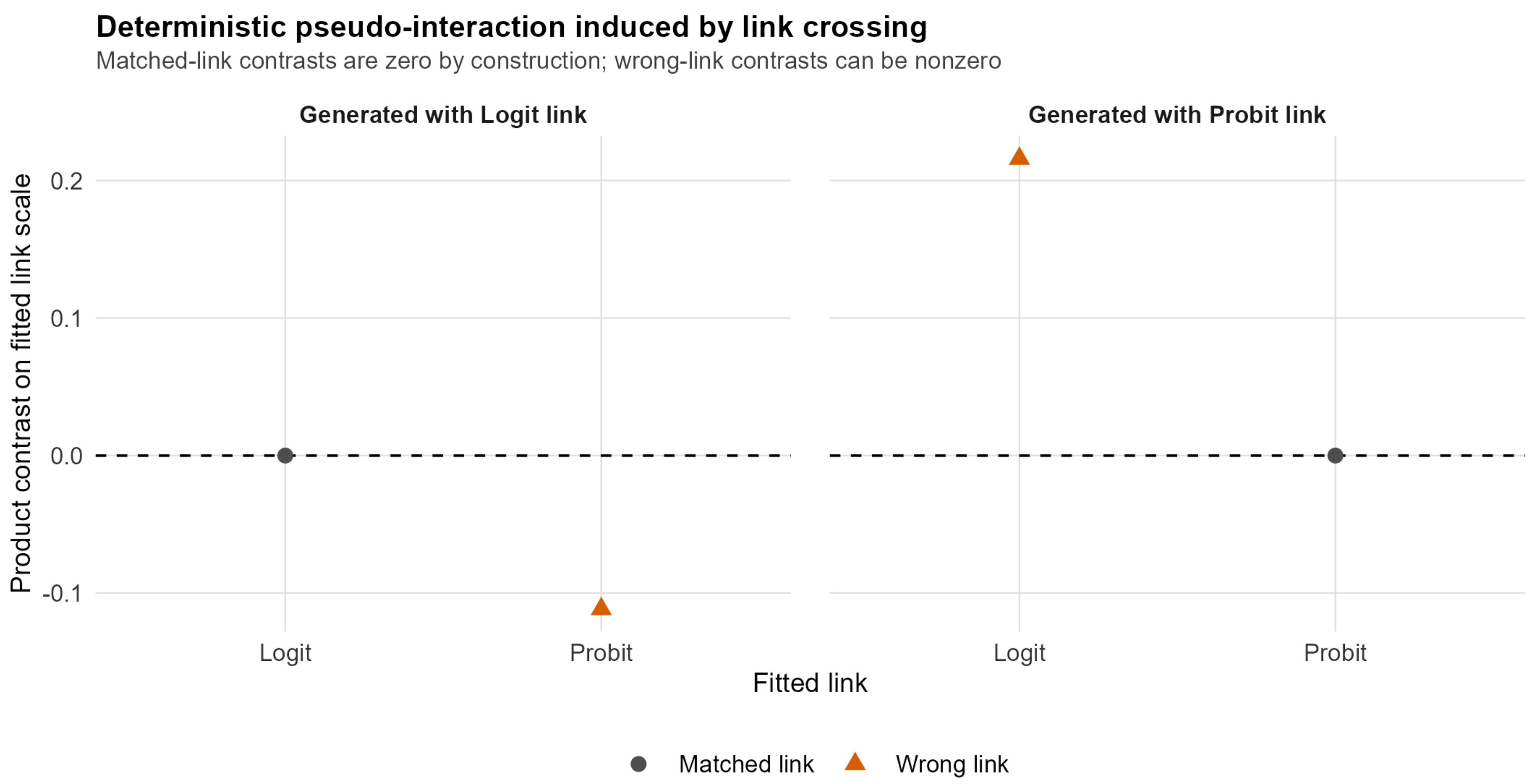
Scenario	Diagnostic	Correct-link flagging	Wrong-link flagging
Gamma mean response time	Pregibon-style added-term link check	3.8% [2.2, 6.7] (287/300)	35.3% [30.1, 40.9] (300/300)

Deterministic pseudo-interaction under the wrong link

Figure 2 shows the deterministic pseudo-interaction induced on the wrong link scale in Simulation 2, before any sampling noise. It makes explicit why matched-link cells imply an exactly additive structure while wrong-link cells do not.

Figure 2

Deterministic pseudo-interaction induced on the wrong link scale in Simulation 2.



S6. Data, code, and computational environment

All analyses were conducted in R, and the manuscript and this supplement were written in Quarto. Each simulation sets a fixed random seed, and the number of Monte Carlo replications and the significance level are recorded together with the saved results. Exact package versions are pinned with `renv`, which is the source of truth for the computational environment; the key modelling dependencies are `glmmTMB` for the mixed-model simulations and `DHARMa` for the diagnostic checks. The complete data, analysis code, preregistration materials, and generated outputs are archived in the public repository associated with the article, so that every table and figure can be regenerated from the raw inputs.

The manuscript simulations use a small number of transparent scenarios. A broader precomputed sensitivity atlas generated in `simulation-atlas/` varies key parameters around those anchors, and the Shiny app explores its compact summaries without running simulations interactively. Replication-level atlas results are intended for the public repository or OSF archive and are not required to render this supplement.

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